

# Rongfeng Cheng | Medical AI Research

Guangzhou, China

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## Education

### South China University of Technology (SCUT)

Guangzhou, China

*Bachelor of Management in Big Data Management and Application*

2023–2027

Cumulative GPA: 3.46/4.0; average score: 85.18/100 (through Semester 6).

Selected AI coursework: Large Models and Generative AI (93/100), Large Language Models and Prompt Engineering (93/100), Machine Learning (92/100), and Deep Learning (88/100).

## Internship Experience

### AI Research Institute, Guangdong Second Provincial General Hospital

*LLM Development Engineer Intern*

Nov 2024–Present

- Developed a standardized-patient Q&A pipeline from approximately 8,000 cardiovascular case PDFs, covering LLM-based information extraction, entity disambiguation, Neo4j knowledge representation, and hybrid retrieval.
- Explored knowledge-graph paths as structural scaffolds for generating medical question–answer and reasoning-trace data for model fine-tuning, evaluation, and retrieval-augmented generation.
- Led full-stack and clinical-workflow development for the five-person Dingbei Meimei acne and facial-health assessment team; supported its initial release when the hospital's Refractory Acne Diagnosis and Treatment Center opened in July 2026.

### Guangzhou Yunzhan Zhichuang Information Technology Co., Ltd.

*Backend & Algorithm Development Intern*

Jul 2025–Jan 2026

- Designed a UMLS-grounded schema and normalization pipeline for heterogeneous health-examination data extracted from multi-institutional reports, including LLM-assisted schema evolution and terminology mapping.
- Developed backend workflow logic for the Health Navigation Plan within the company's iTwins product.

## Manuscripts & Publications

**Under Review [1]: R. Cheng et al.** “DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification.” *IEEE BIBM 2026*. First author.

**Under Review [2]: R. Cheng et al.** “A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification.” *IEEE JBHI*. First author.

**Major Revision [3]:** “Implicit Medical Ethical Deficits in Large Language Models: A Benchmarking and Alignment Study.” Submitted to *Nature Communications*. Sixth author; leading the revision-stage experiments for the response to reviewers.

**Ongoing: PathCLIP: Evidence-Aware ECG–Text Representation Learning.** Ongoing first-author study of ECG–text alignment and zero-shot recognition with matched-count, permuted-hierarchy, residual-only, and held-out-label controls.

**Apr 2026 [4]: J. He, S. Qin, H. Zou, F. Jing, W. Liu, C. Li, Q. Wang, Z. Hu, K. Song, Z. Xu, Q. Luo, Y. Cheng, R. Cheng, H. Zhang, and W. Cheng.** “CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation.” *iScience*. doi:10.1016/j.isci.2026.115073.

## Research & Project Experience

*Institute of Healthcare Artificial Intelligence Application, Guangdong Second Provincial General Hospital*

### DLEF: Dataset-Label Evidence Fingerprints for Explaining Cross-Dataset Performance in ECG Classification

Jun 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Led study design, implementation, experiments, analysis, and manuscript preparation; proposed directional evidence fingerprints for characterizing dataset-label transfer risk.
- Across CODE-15%, PTB-XL, and Chapman-Shaoxing, DLEF distance was associated with label-level external AUROC degradation (Spearman  $r = 0.843$  across 108 points;  $r = 0.892$  after source-target-label aggregation).

### A Matched-Control Perturbation Audit Framework for Evaluating Evidence Use in Deep Learning Models: An Application to 12-Lead ECG Classification

Jan 2026–Present

Advisors: Haobin Zhang and Weibin Cheng.

- Led study design, implementation, experiments, analysis, and manuscript preparation; developed a matched-control perturbation audit spanning clinically defined evidence units, operator comparisons, and frozen-weight external transfer.
- Evaluated five diagnoses and three raw-signal models on CODE-15% and PTB-XL; the audit classified 6 of 15 model-diagnosis pairs as evidence-supported and showed that matched controls or operator choice can overturn target-only conclusions.

*Co-founder & Core Developer*

- Co-developed a cardiovascular health-management platform integrating PPG/ECG data, intelligent consultation, risk assessment, and proactive health workflows.
- Led model fine-tuning, RAG, and full-stack implementation using LoRA, DPO, MQ-RAG, and chain-of-thought methods; also prepared grant and competition application materials.
- Deployed the initial mini-program at Guangdong Second Provincial General Hospital and currently leads full-stack development of version 2.0; supported by national-level innovation-program projects in 2025 and 2026 and incorporated as Yujian Future Mind Technology (Zhongshan) Co., Ltd. in Aug 2026.

**Myocardial Ischemia Knowledge Graph**

2025

*Project Lead, SCUT Student Research Program*

Advisor: Xunde Dong. Led a five-student interdisciplinary team; designed the ontology, annotation schema, guidelines, and quality-control workflow; built a multi-annotator platform with adjudication and agreement checks; and implemented XGBoost-based implicit reasoning with graph-mapped interpretability analysis.

## Honors & Awards

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**2025:** Campus Gold Award, Internet+ Competition – Dingbei Xinxin cardiovascular health-management project.

**2025:** Second Prize, 9th Guangzhou Mathematical Modeling University League – fruit classification with ConvNeXt.

**2025:** Third Prize, China Undergraduate Mathematical Contest in Modeling, Guangdong Division – nonlinear programming, genetic algorithms, and particle-swarm optimization.

## Patents & Software IP

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**Patent: Method, System, and Electronic Device for Medical Question-Answering Output Assurance** – First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.

**Patent: Method and System for Integrative Chinese–Western Medicine-Assisted Acne Assessment Based on Multimodal Data Fusion** – First Inventor; application submitted Jul 2026; examination documents pending.

**Software IP:** Knowledge Graph Ontology Intelligent Management System V1.0 (#2025SR2454864); LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 (#2025SR2080673).

## Skills

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**Modeling:** PyTorch, XGBoost, representation learning, multimodal learning, LLM fine-tuning, optimization

**Evaluation & Data Curation:** RAG/retrieval evaluation, perturbation analysis, external validation, schema design

**LLM & Knowledge Systems:** LoRA, DPO, RAG/MQ-RAG, knowledge graphs, information extraction, UMLS

**Engineering:** Python, TypeScript, SQL, Neo4j, Flask, Git

## Languages

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Chinese (native)   Cantonese (native)   English: IELTS 6.5, CET-6 528